PIZZA SALES ANALYSIS

SQL Data Analysis Project

MySQL Workbench | Database: pizzahut

13 Analytical Queries | 4 Tables | Advanced SQL

PROJECT OVERVIEW

4
Total Tables

13
SQL Questions

32
Pizza Types

pizzahut
Database

About This Project

This project performs end-to-end SQL analysis on a pizza restaurant's sales data using MySQL Workbench.
The database 'pizzahut' contains four tables: **orders, order_details, pizzas, and pizza_types.**
Using JOIN operations, aggregate functions, window functions (RANK), CTEs, and subqueries, 13 business questions were answered to derive actionable insights on revenue, popular pizzas, ordering patterns, and category performance.

DATABASE SCHEMA

ORDERS

- order_id (PK)
- order_date
- order_time

PIZZAS

- pizza_id (PK)
- pizza_type_id (FK)
- size
- price

ORDER_DETAILS

- order_details_id (PK)
- order_id (FK)
- pizza_id (FK)
- quantity

PIZZA_TYPES

- pizza_type_id (PK)
- name
- category
- ingredients

Relationships: orders ←→ order_details (order_id) | order_details ←→ pizzas (pizza_id) | pizzas ←→ pizza_types (pizza_type_id)

BASIC LEVEL QUERIES

Q1–Q4

Q1: Total Orders

```
SELECT COUNT(order_id) AS Total_orders FROM orders;
```

Retrieves the total number of pizza orders placed

Q2: Total Revenue

```
SELECT ROUND(SUM(quantity * price), 3) AS Total_Revenue
FROM orders_details JOIN pizzas ON ...
```

Calculates total revenue generated from all pizza sales

Q3: Highest-Priced Pizza

```
SELECT name, price FROM pizza_types JOIN pizzas
ORDER BY price DESC LIMIT 1;
```

Identifies the most expensive pizza on the menu

Q4: Most Common Size

```
SELECT size, COUNT(order_details_id) AS order_count
FROM pizzas JOIN orders_details GROUP BY size;
```

Finds the pizza size ordered most frequently

INTERMEDIATE LEVEL QUERIES

Q5–Q8

Q5: Top 5 Most Ordered Pizzas

```
SELECT name, SUM(quantity) AS most_ordered
FROM pizza_types JOIN pizzas JOIN orders_details
GROUP BY name ORDER BY most_ordered DESC LIMIT 5;
```

→ *BBQ Chicken (1985), Pepperoni (1967), Classic Deluxe (1942)...*

Q6: Quantity by Category

```
SELECT category, SUM(quantity) AS Total_quantity
FROM pizza_types JOIN pizzas JOIN orders_details
GROUP BY category ORDER BY Total_quantity DESC;
```

→ *Classic (11984) > Supreme (9671) > Veggie (9423) > Chicken (8869)*

Q7: Orders by Hour of Day

```
SELECT HOUR(order_time) AS hour,
COUNT(order_id) AS order_count
FROM orders GROUP BY HOUR(order_time);
```

→ *Peak hours: 12 PM (2520 orders), 1 PM (2455), 6 PM (1920)*

Q8: Category-wise Pizza Distribution

```
SELECT category, COUNT(name) AS pizza_names
FROM pizza_types GROUP BY category;
```

→ *Chicken (6), Classic (8), Supreme (9), Veggie (9)*

ADVANCED LEVEL QUERIES

Q9–Q13

Q9: Average Pizzas Ordered Per Day (CTE)

```
WITH order_quantity AS (
  SELECT order_date, SUM(quantity) AS Total_quantity
  FROM orders JOIN orders_details ON ...
  GROUP BY order_date
)
SELECT ROUND(AVG(total_quantity),0) AS order_per_day FROM order_quantity;
```

Result

132 pizzas ordered per day on average

Uses CTE + AVG() + ROUND() functions

Q10: Top 3 Pizzas by Revenue per Category

```
SELECT name, revenue FROM
  (SELECT category, name, revenue,
   RANK() OVER(PARTITION BY category
   ORDER BY revenue DESC) AS ranks
   FROM (...) AS pizza_category) AS b
WHERE ranks <= 3;
```

Result (Top per Category)

BBQ Chicken Pizza: $34,912
Thai Chicken Pizza: $34,440
Cali Chicken Pizza: $33,203

ADVANCED LEVEL QUERIES (CONT.)

Q11–Q13

Q11: % Revenue Contribution by Category

```
SELECT category,
  ROUND(SUM(quantity*price)/(SELECT SUM(quantity*price)
  FROM orders_details JOIN pizzas)*100,2)
  AS revenue FROM ...
```

→ Classic: 26.9% | Supreme: 25.51%
Chicken: 23.83% | Veggie: 23.76%

Q12: Cumulative Revenue Over Time

```
SELECT order_date,
  SUM(Total_Revenue) OVER(
  ORDER BY order_date)
  AS cumulative_revenue FROM (SELECT ...);
```

→ 2015-01-01: $3,344.8
2015-01-02: $6,463.95 ... (growing daily)

Q13: Top 3 Revenue per Category (RANK)

```
RANK() OVER(PARTITION BY category
ORDER BY revenue DESC)
WHERE ranks <= 3;
```

→ 12 rows returned — top 3 earners identified per pizza category

KEY INSIGHTS & FINDINGS

#1 Best-Selling Pizza

The Barbecue Chicken Pizza leads in both quantity (1,985 orders) and revenue ($34,912.75)

Peak Ordering Time

Lunch hours dominate — 12 PM (2,520) and 1 PM (2,455) are the busiest ordering times

Top Category

Classic pizzas are the most ordered category with 11,984 total quantity sold

Revenue Leader

Classic category contributes 26.9% of total revenue, closely followed by Supreme at 25.51%

Preferred Size

Large (L) size is the most popular choice among customers based on order count

Daily Average

On average, 132 pizzas are ordered per day across all categories and sizes

SQL CONCEPTS & TECHNIQUES USED

JOINs

Multi-table JOINs across 3-4 tables simultaneously to combine pizza, order, and type data

Aggregate Functions

SUM(), COUNT(), AVG(), ROUND() for revenue and order calculations

Window Functions

RANK() OVER (PARTITION BY) to rank pizzas within each category

CTEs (WITH clause)

Common Table Expressions for multi-step calculations like average orders/day

Subqueries

Nested SELECT statements for percentage revenue and comparative analysis

Running Totals

SUM() OVER (ORDER BY) for cumulative revenue trend analysis over time

Tool: MySQL Workbench 8.0 | Language: SQL (MySQL) | Database: pizzahut

THANK YOU!

Pizza Sales SQL Analysis Project

GitHub

https://github.com/sahidul-islam786/pizza-sales-SQL-project

LinkedIn

www.linkedin.com/in/mohammad-saidul-islam-6706b1272

Built with MySQL Workbench | 13 SQL Queries | 4 Tables | Real-world Data Analysis